Que. Expand the van der Waals equation of state as a cubic equation of volume. At the critical point, the three roots of this equation must collapse to a single root, i.e., $(V-V_C)^3 = 0$. Compare these two equations term-by-term and evaluate the critical volume, pressure, and temperature in terms of the constants a, b, and R. Now, define the reduced volume/pressure/temperature as the volume/pressure/temperature divided by the critical volume/pressure/temperature and re-write the van der Waals equation in terms of reduced parameters. What is the importance of this equation?

1. Van der Waals equation as a cubic in volume

The van der Waals equation (per mole) is

$$\left(P + \frac{a}{V^2}\right)(V-b) = RT$$

Expand:

$$PV - Pb + \frac{a}{V} - \frac{ab}{V^2} = RT$$

Multiply throughout by V^2:

$$PV^3 - PbV^2 + aV - ab = RTV^2$$

Rearranging,

$$\boxed{PV^3 - (Pb+RT)V^2 + aV - ab =}$$

This is a **cubic equation in** V.

2. Condition at the critical point

At the critical point (P_c, T_c, V_c), the three roots of the cubic are equal:

$$(V-V_c)^3 = 0$$

Expanding,

$$V^3 - 3V_cV^2 + 3V_c^2V - V_c^3 = 0$$

Divide the cubic equation by P and compare with the above form:

$$V^3 - \frac{(Pb+RT)}{P}V^2 + \frac{a}{P}V - \frac{ab}{P} = 0$$

3. Comparison of coefficients

Matching coefficients term by term at the critical point:

$$3V_c = \frac{P_cb + RT_c}{P_c} \quad (1)$$

$$3V_c^2 = \frac{a}{P_c} \quad (2)$$

$$V_c^3 = \frac{ab}{P_c} \quad (3)$$

(a) Critical volume

From (2) and (3):

$$\frac{a}{3V_c^2} = \frac{ab}{V_c^3}$$

$$\Rightarrow V_c = 3b$$

(b) Critical pressure

Substitute $V_c = 3b$ into (2):

$$P_c = \frac{a}{3(3b)^2} = \frac{a}{27b^2}$$

(c) Critical temperature

Using equation (1):

$$3(3b) = b + \frac{RT_c}{P_c}$$

$$T_c = \frac{8a}{27Rb}$$

$$\boxed{\begin{aligned} V_c &= 3b \\ P_c &= \frac{a}{27b^2} \\ T_c &= \frac{8a}{27Rb} \end{aligned}}$$

5. Reduced variables

Define reduced (dimensionless) quantities:

$$V_r = \frac{V}{V_c}, \quad P_r = \frac{P}{P_c}, \quad T_r = \frac{T}{T_c}$$

Substitute

$$V = 3bV_r, \quad P = \frac{a}{27b^2}P_r, \quad T = \frac{8a}{27Rb}T_r$$

into the van der Waals equation.

After simplification:

$$\boxed{\left(P_r + \frac{3}{V_r^2}\right)(3V_r - 1) = 8T_r}$$

6. Importance of the reduced equation

1. It is **independent of substance-specific constants** a and b.
2. It demonstrates the **Law of Corresponding States**.
3. All fluids obeying the van der Waals equation behave identically when expressed in reduced variables.
4. Explains universal behavior of gases near the **critical point**.
5. Useful for comparing experimental data of different gases on a common scale.

E1C.9(a) A certain gas obeys the van der Waals equation with $a = 0.50\,\text{m}^6\,\text{Pa}\,\text{mol}^{-2}$. Its molar volume is found to be $5.00 \times 10^{-4}\,\text{m}^3\,\text{mol}^{-1}$ at 273 K and 3.0 MPa. From this information calculate the van der Waals constant b. What is the compression factor for this gas at the prevailing temperature and pressure?

E1C.9(a) The van der Waals equation of state in terms of the molar volume is given by [1C.5b–24], $p = RT/(V_\text{m} - b) - a/V_\text{m}^2$. This relationship is rearranged to find b

$$p = \frac{RT}{V_\text{m} - b} - \frac{a}{V_\text{m}^2} \quad \text{hence} \quad p + \frac{a}{V_\text{m}^2} = \frac{RT}{V_\text{m} - b}$$

$$\text{hence} \quad \frac{pV_\text{m}^2 + a}{V_\text{m}^2} = \frac{RT}{V_\text{m} - b} \quad \text{hence} \quad \frac{V_\text{m}^2}{pV_\text{m}^2 + a} = \frac{V_\text{m} - b}{RT}$$

$$\text{hence} \quad b = V_\text{m} - \frac{RTV_\text{m}^2}{pV_\text{m}^2 + a}$$

With the data given

$$b = V_\text{m} - \frac{RTV_\text{m}^2}{pV_\text{m}^2 + a} = (5.00 \times 10^{-4}\,\text{m}^3\,\text{mol}^{-1})$$

$$- \frac{(8.3145\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}) \times (273\,\text{K}) \times (5.00 \times 10^{-4}\,\text{m}^3\,\text{mol}\;\;,}{(3.0 \times 10^6\,\text{Pa}) \times (5.00 \times 10^{-4}\,\text{m}^3\,\text{mol}^{-1})^2 + (0.50\,\text{m}^6\,\text{Pa}\,\text{mol}^{-2})}$$

$$= \boxed{4.6 \times 10^{-5}\,\text{m}^3\,\text{mol}^{-1}}$$

where $1\,\text{Pa} = 1\,\text{kg}\,\text{m}^{-1}\,\text{s}^{-2}$ and $1\,\text{J} = 1\,\text{kg}\,\text{m}^2\,\text{s}^{-2}$ have been used.

The compression factor Z is defined in [1C.1–20] as $Z = V_\text{m}/V_\text{m}^\circ$, where V_m° is the molar volume of a perfect gas under the same conditions. This volume is computed from the equation of state for a perfect gas, [1A.4–8], as $V_\text{m}^\circ = RT/p$, hence $Z = pV_\text{m}/RT$, [1C.2–20]. With the data given

$$Z = \frac{pV_\text{m}}{RT} = \frac{(3.0 \times 10^6\,\text{Pa}) \times (5.00 \times 10^{-4}\,\text{m}^3\,\text{mol}^{-1})}{(8.3145\,\text{J}\,\text{K}^{-1}\,\text{mol}^{-1}) \times (273\,\text{K})} = \boxed{0.66}$$

E2A.4(a) A sample consisting of 1.00 mol Ar is expanded isothermally at 20 °C from 10.0 dm^3 to 30.0 dm^3 (i) reversibly, (ii) against a constant external pressure equal to the final pressure of the gas, and (iii) freely (against zero external pressure). For the three processes calculate q, w, and ΔU.

E2A.4(a) For all cases $\Delta U = 0$, because the internal energy of a perfect gas depends on the temperature alone.

(i) The work of reversible isothermal expansion of a perfect gas is given by [2A.9–41]

$$w = -nRT\ln\left(\frac{V_f}{V_i}\right)$$

$$= -(1.00\ \text{mol}) \times (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (293.15\ \text{K}) \times \ln\left(\frac{30.0\ \text{dm}^3}{10.0\ \text{dm}^3}\right)$$

$$= -2.68 \times 10^3\ \text{J} = \boxed{-2.68\ \text{kJ}}$$

Note that the temperature is expressed in K in the above equation. Using the First Law of thermodynamics, [2A.2–38], gives

$$q = \Delta U - w = 0 - (-2.68\ \text{kJ}) = \boxed{+2.68\ \text{kJ}}$$

(ii) The final pressure of the expanding gas is found using the perfect gas law, [1A.4–8]

$$p_f = \frac{nRT}{V_f} = \frac{(1.00\ \text{mol}) \times (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (293.15\ \text{K})}{(30.0 \times 10^{-3}\ \text{m}^3)}$$

$$= 8.12... \times 10^4\ \text{Pa}$$

This pressure equals the constant external pressure against which the gas is expanding, therefore the work of expansion is

$$w = -p_{ex} \times \Delta V = (8.12... \times 10^4\ \text{Pa}) \times (30.0 \times 10^{-3}\ \text{m}^3 - 10.0 \times 10^{-3}\ \text{m}^3)$$

$$= -1.62 \times 10^3\ \text{J} = \boxed{-1.62\ \text{kJ}}$$

and hence $q = \boxed{+1.62\ \text{kJ}}$

(iii) Free expansion is expansion against zero force, so $\boxed{w = 0}$ and therefore $\boxed{q = 0}$ as well.

E2A.5(a) A sample consisting of 1.00 mol of perfect gas atoms, for which $C_{V,\mathrm{m}} = \frac{3}{2}R$, initially at $p_1 = 1.00\,\mathrm{atm}$ and $T_1 = 300\,\mathrm{K}$, is heated reversibly to 400 K at constant volume. Calculate the final pressure, ΔU, q, and w.

E2A.5(a) For a perfect gas at constant volume $p_\mathrm{i}/T_\mathrm{i} = p_\mathrm{f}/T_\mathrm{f}$ therefore,

$$p_\mathrm{f} = p_\mathrm{i} \times \frac{T_\mathrm{f}}{T_\mathrm{i}} = (1.00\ \mathrm{atm}) \times \left(\frac{400\ \mathrm{K}}{300\ \mathrm{K}}\right) = \boxed{1.33\ \mathrm{atm}}$$

The change in internal energy at constant volume is given by [2A.15b–45]

$$\Delta U = nC_{V,\mathrm{m}}\Delta T = (1.00\ \mathrm{mol}) \times \left(\frac{3}{2} \times 8.3145\ \mathrm{J\,K^{-1}\,mol^{-1}}\right) \times (400\ \mathrm{K} - 300\ \mathrm{K})$$

$$= +1.25 \times 10^3\ \mathrm{J} = \boxed{+1.25\ \mathrm{kJ}}$$

The volume of the gas is constant, so the work of expansion is zero, $\boxed{w = 0}$. The First Law of thermodynamics gives $q = \Delta U - w = +1.25\ \mathrm{kJ} - 0 = \boxed{+1.25\ \mathrm{kJ}}$.

E2B.4(a) When 3.0 mol O_2 is heated at a constant pressure of 3.25 atm, its temperature increases from 260 K to 285 K. Given that the molar heat capacity of O_2 at constant pressure is 29.4 J K^{-1} mol^{-1}, calculate q, ΔH, and ΔU.

E2B.4(a) Under constant pressure $q_p = \Delta H$, therefore

$$q_p = \Delta H = nC_{p,m}\Delta T = (3.0\ \text{mol}) \times (29.4\ \text{J K mol}^{-1}) \times (285\ \text{K} - 260\ \text{K})$$
$$= 2.20...\ \text{kJ} = \boxed{+2.2\ \text{kJ}}$$

The definition of enthalpy is given by [2B.1–46], $H = U + pV$. For a change at constant pressure, it follows that $\Delta H = \Delta U + p\Delta V$, where $\Delta V = V_f - V_i$.

If the gas is assumed to be perfect, then $V_f = nRT_f/p$ and $V_i = nRT_i/p$, so $\Delta V = (T_f - T_i)nR/p$. Hence $p\Delta V = nR(T_f - T_i) = nR\Delta T$.

$$\Delta U = \Delta H - nR\Delta T$$
$$= (2.20...\ \text{kJ}) - (3.0\ \text{mol}) \times (8.3145\ \text{J K}^{-1}\ \text{mol}^{-1}) \times (285\ \text{K} - 260\ \text{K})$$
$$= \boxed{+1.6\ \text{kJ}}$$

P2D.2 Starting from the expression $C_p - C_V = T(\partial p/\partial T)_V(\partial V/\partial T)_p$, use the appropriate relations between partial derivatives (*The chemist's toolkit* 9 in Topic 2A) to show that

$$C_p - C_V = \frac{T(\partial V/\partial T)_p^2}{(\partial V/\partial p)_T}$$

Use this expression to evaluate $C_p - C_V$ for a perfect gas.

We start from the given thermodynamic identity

$$\boxed{C_P - C_V = T\left(\frac{\partial p}{\partial T}\right)_V\left(\frac{\partial V}{\partial T}\right)_p}$$

1. Use partial-derivative relations (chemist's toolkit)

A key cyclic (reciprocity) relation between partial derivatives is

$$\boxed{\left(\frac{\partial V}{\partial T}\right)_p = -\frac{\left(\frac{\partial p}{\partial T}\right)_V}{\left(\frac{\partial p}{\partial V}\right)_T}}$$

This follows from the general identity

$$\left(\frac{\partial x}{\partial y}\right)_z\left(\frac{\partial y}{\partial z}\right)_x\left(\frac{\partial z}{\partial x}\right)_y = -1$$

2. Substitute into the heat-capacity expression

Insert (2) into (1):

$$C_P - C_V = T\left(\frac{\partial p}{\partial T}\right)_V\left[-\frac{\left(\frac{\partial p}{\partial T}\right)_V}{\left(\frac{\partial p}{\partial V}\right)_T}\right]$$

$$\boxed{C_P - C_V = -T\frac{\left(\frac{\partial p}{\partial T}\right)_V^2}{\left(\frac{\partial p}{\partial V}\right)_T}}$$

This is the required result:

$$\boxed{C_P - C_V = -T\frac{(\partial p/\partial T)_V^2}{(\partial p/\partial V)_T}}$$

3. Evaluate $C_P - C_V$ for a perfect (ideal) gas

For one mole of an ideal gas:

$$pV = RT \quad \Rightarrow \quad p = \frac{RT}{V}$$

Required derivatives

$$\left(\frac{\partial p}{\partial T}\right)_V = \frac{R}{V}$$

$$\left(\frac{\partial p}{\partial V}\right)_T = -\frac{RT}{V^2}$$

4. Substitute into the general expression

$$C_P - C_V = -T\frac{\left(\frac{R}{V}\right)^2}{-\frac{RT}{V^2}}$$

Simplify:

$$C_P - C_V = R$$

5. Final result

$$\boxed{C_P - C_V = R}$$

2.20 Show that the following functions have exact differentials: (a) $x^2y + 3y^2$, (b) $x \cos xy$, (c) x^3y^2, (d) $t(t + e^s) + s$.

(a) $f(x, y) = x^2y + 3y^2$

Compute the differential:

$$df = \left(\frac{\partial f}{\partial x}\right) dx + \left(\frac{\partial f}{\partial y}\right) dy$$

$$\frac{\partial f}{\partial x} = 2xy \quad \Rightarrow \quad M = 2xy$$

$$\frac{\partial f}{\partial y} = x^2 + 6y \quad \Rightarrow \quad N = x^2 + 6y$$

Check exactness:

$$\frac{\partial M}{\partial y} = 2x \quad , \quad \frac{\partial N}{\partial x} = 2x$$

Since they are equal,

$$\boxed{df \text{ is exact}}$$

(b) $f(x, y) = x\cos(xy)$

$$\frac{\partial f}{\partial x} = \cos(xy) - xy\sin(xy) \Rightarrow M = \cos(xy) - xy\sin(xy)$$

$$\frac{\partial f}{\partial y} = -x^2\sin(xy) \Rightarrow N = -x^2\sin(xy)$$

Now,

$$\frac{\partial M}{\partial y} = -x\sin(xy) - x^2y\cos(xy)$$

$$\frac{\partial N}{\partial x} = -2x\sin(xy) - x^2y\cos(xy)$$

These are **not equal**, so

$$\boxed{df \text{ is NOT exact}}$$

(c) $f(x, y, z) = x^3y^2$

Total differential:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

$$\frac{\partial f}{\partial x} = 3x^2y^2, \quad \frac{\partial f}{\partial y} = 2x^3y, \quad \frac{\partial f}{\partial z} = 0$$

Since f is a well-defined scalar function, its total differential is **automatically exact**.

$$\boxed{df \text{ is exact}}$$

(d) $f(t, s) = t(t + e^s) + s$

Simplify:

$$f = t^2 + te^s + s$$

$$\frac{\partial f}{\partial t} = 2t + e^s \Rightarrow M = 2t + e^s$$

$$\frac{\partial f}{\partial s} = te^s + 1 \Rightarrow N = te^s + 1$$

Check:

$$\frac{\partial M}{\partial s} = e^s \quad , \quad \frac{\partial N}{\partial t} = e^s$$

They are equal, hence

$$\boxed{df \text{ is exact}}$$

E2E.1(a) Use the equipartition principle to estimate the values of $\gamma = C_p/C_V$ for gaseous ammonia and methane. Do this calculation with and without the vibrational contribution to the energy. Which is closer to the experimental value at 25 °C?

The experimental values of γ for ammonia and methane are the same, $\gamma = 1.31$,

Key theory: Equipartition principle

Each quadratic degree of freedom contributes:

- $\frac{1}{2}R$ to C_V
- Translational: **3**
- Rotational:
  - Linear molecule: **2**
  - Non-linear molecule: **3**
- Vibrational mode:
  - Each vibrational mode contributes **R** to C_V (½R kinetic + ½F

Also:

$$C_P = C_V + R$$

Degrees of freedom

- Translational = 3
- Rotational = 3
- Vibrational modes:

$$3N - 6 = 3(5) - 6 = 9$$

(a) Without vibration

$$C_V = \left(\frac{3}{2} + \frac{3}{2}\right)R = 3R$$

$$C_P = 4R$$

$$\boxed{\gamma = \frac{4R}{3R} = 1.33}$$

(b) With vibration

Vibrational contribution:

$$9R$$

$$C_V = 3R + 9R = 12R$$

$$C_P = 13R$$

$$\boxed{\gamma = \frac{13}{12} = 1.08}$$

2 Ammonia (NH_3)

NH_3 is also a **non-linear polyatomic molecule.**

Degrees of freedom

- Translational = 3
- Rotational = 3
- Vibrational modes:

$$3N - 6 = 3(4) - 6 = 6$$

(a) Without vibration

$$C_V = 3R$$

$$C_P = 4R$$

$$\boxed{\gamma = 1.33}$$

(b) With vibration

Vibrational contribution:

$$6R$$

$$C_V = 3R + 6R = 9R$$

$$C_P = 10R$$

$$\boxed{\gamma = \frac{10}{9} = 1.11}$$

I2.5 As shown in *The chemist's toolkit* 9 in Topic 2A, it is a property of partial derivatives that

$$\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = \left(\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x\right)_y$$

Use this property and eqn 2A.14 to write an expression for $(\partial C_V/\partial V)_T$ as a second derivative of U and find its relation to $(\partial U/\partial V)_T$. Then show that $(\partial C_V/\partial V)_T = 0$ for a perfect gas.

Given property of partial derivatives

For a well-behaved function $f(x, y)$,

$$\left(\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y\right)_x = \left(\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x\right)_y$$

Step 1: Start from the definition of C_V

From thermodynamics (Eqn 2A.14 in Atkins):

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Differentiate both sides with respect to V at constant T:

$$\left(\frac{\partial C_V}{\partial V}\right)_T = \left(\frac{\partial}{\partial V}\left(\frac{\partial U}{\partial T}\right)_V\right)_T$$

Step 2: Use the property of mixed partial derivatives

Using the given property:

$$\left(\frac{\partial}{\partial V}\left(\frac{\partial U}{\partial T}\right)_V\right)_T = \left(\frac{\partial}{\partial T}\left(\frac{\partial U}{\partial V}\right)_T\right)_V$$

Hence,

$$\boxed{\left(\frac{\partial C_V}{\partial V}\right)_T = \left(\frac{\partial}{\partial T}\left(\frac{\partial U}{\partial V}\right)_T\right)_V}$$

This expresses $(\partial C_V/\partial V)_T$ as a **second derivative of internal energy**.

Step 3: Apply this result to a perfect gas

For a **perfect (ideal) gas**:

$$U = U(T) \quad \text{only}$$

Therefore,

$$\left(\frac{\partial U}{\partial V}\right)_T = 0$$

Now differentiate with respect to T:

$$\left(\frac{\partial}{\partial T}\left(\frac{\partial U}{\partial V}\right)_T\right)_V = 0$$

Step 4: Final result

$$\boxed{\left(\frac{\partial C_V}{\partial V}\right)_T = 0 \quad \text{for a perfect gas}}$$

E3A.2(a) Consider a process in which 100 kJ of energy is transferred reversibly and isothermally as heat to a large block of copper. Calculate the change in entropy of the block if the process takes place at (i) 0 °C, (ii) 50 °C.

E3A.2(a) The thermodynamic definition of entropy is [3A.1a–*80*], $dS = dq_{rev}/T$ or for a finite change at constant temperature $\Delta S = q_{rev}/T$. The transfer of heat is specified as being reversible, which can often be assumed for a large enough metal block, therefore $q_{rev} = 100$ kJ.

(i)

$$\Delta S = \frac{q_{rev}}{T} = \frac{100\ \text{kJ}}{273.15\ \text{K}} = 0.366\ \text{kJ} = \boxed{+366\ \text{J}}$$

(ii)

$$\Delta S = \frac{q_{rev}}{T} = \frac{100\ \text{kJ}}{(273.15\ \text{K} + 50\ \text{K})} = 0.309\ \text{kJ} = \boxed{+309\ \text{J}}$$

E3A.5(a) In a certain ideal heat engine, 10.00 kJ of heat is withdrawn from the hot source at 273 K and 3.00 kJ of work is generated. What is the temperature of the cold sink?

1. Known quantities for an ideal heat engine (Carnot):

- Heat from hot source: $Q_H = 10.00\ \mathrm{kJ}$
- Work output: $W = 3.00\ \mathrm{kJ}$
- Temperature of hot source: $T_H = 273\ \mathrm{K}$
- Cold sink temperature: $T_C = ?$

---

2. Efficiency relations for a Carnot engine

Ideal heat engine = Carnot engine → efficiency:

$$\eta = \frac{W}{Q_H} = 1 - \frac{T_C}{T_H}$$

3. Substitute given numbers

$$\frac{3.00}{10.00} = 0.3000$$

$$0.3000 = 1 - \frac{T_C}{273}$$

$$\frac{T_C}{273} = 1 - 0.3000 = 0.7000$$

$$T_C = 273 \times 0.7000 = 191.1\ \mathrm{K}$$

E3B.7(a) Calculate the change in entropy of the system when 10.0 g of ice at −10.0 °C is converted into water vapour at 115.0 °C and at a constant pressure of 1 bar. The molar constant-pressure heat capacities are: $C_{p,m}(H_2O(s)) = 37.6\ J\ K^{-1}\ mol^{-1}$; $C_{p,m}(H_2O(l)) = 75.3\ J\ K^{-1}\ mol^{-1}$; and $C_{p,m}(H_2O(g)) = 33.6\ J\ K^{-1}\ mol^{-1}$. The standard enthalpy of vaporization of $H_2O(l)$ is $40.7\ kJ\ mol^{-1}$, and the standard enthalpy of fusion of $H_2O(l)$ is $6.01\ kJ\ mol^{-1}$, both at the relevant transition temperatures.

- Mass $m = 10.0$ g of H_2O
- Molar mass $M = 18.015$ g/mol → $n = \frac{10.0}{18.015} \approx 0.555$ mol
- Initial: ice at $T_1 = -10.0\ ^\circ C = 263.15$ K
- Final: vapor at $T_2 = 115.0\ ^\circ C = 388.15$ K, constant $P = 1$ bar

Heat capacities:

$$C_{p,m}(s) = 37.6\ J\ mol^{-1}\ K^{-1}$$

$$C_{p,m}(l) = 75.3\ J\ mol^{-1}\ K^{-1}$$

$$C_{p,m}(g) = 33.6\ J\ mol^{-1}\ K^{-1}$$

Phase transitions:

Melting point $T_m = 0.00\ ^\circ C = 273.15$ K, $\Delta_{fus}H = 6.01$ kJ/mol

Boiling point $T_b = 100.00\ ^\circ C = 373.15$ K, $\Delta_{vap}H = 40.7$ kJ/mol

2. Process steps

1. Warm ice from 263.15 K to 273.15 K
2. Melt ice at 273.15 K
3. Warm liquid from 273.15 K to 373.15 K
4. Vaporize liquid at 373.15 K
5. Warm vapor from 373.15 K to 388.15 K

Each step is reversible for entropy calculation.

3. Entropy changes

$$\Delta S = n\left[C_{p,m}(s)\ln\frac{T_m}{T_1} + \frac{\Delta_{fus}H}{T_m} + C_{p,m}(l)\ln\frac{T_b}{T_m} + \frac{\Delta_{vap}H}{T_b} + C_{p,m}(g)\ln\frac{T_2}{T_b}\right]$$

Step 1: Warming ice

$$\Delta S_1 = n \times 37.6 \times \ln\frac{273.15}{263.15}$$

$$\ln\frac{273.15}{263.15} \approx \ln(1.03800) \approx 0.03731$$

$$\Delta S_1 \approx 0.555 \times 37.6 \times 0.03731 \approx 0.555 \times 1.402 \approx 0.778\ J/K$$

Step 2: Melting

$$\Delta S_2 = n \times \frac{6010}{273.15}$$

$$\frac{6010}{273.15} \approx 22.$$

$$\Delta S_2 \approx 0.555 \times 22.00 \approx 12.21\ J/K$$

Step 3: Warming liquid

$$\Delta S_3 = n \times 75.3 \times \ln\frac{373.15}{273.15}$$

$$\ln\frac{373.15}{273.15} \approx \ln(1.3660) \approx 0.3118$$

$$\Delta S_3 \approx 0.555 \times 75.3 \times 0.3118 \approx 0.555 \times 23.48 \approx 13.03\ J/K$$

Step 4: Vaporization

$$\Delta S_4 = n \times \frac{40700}{373.15}$$

$$\frac{40700}{373.15} \approx 109.07$$

$$\Delta S_4 \approx 0.555 \times 109.07 \approx 60.53\ J/K$$

Step 5: Warming vapor

$$\Delta S_5 = n \times 33.6 \times \ln\frac{388.15}{373.15}$$

$$\ln\frac{388.15}{373.15} \approx \ln(1.04020) \approx 0.03942$$

4. Summation

$$\Delta S_{total} \approx 0.778 + 12.21 + 13.03 + 60.53 + 0.735$$

$$= 87.283\ J/K$$

E3C.1(a) At 4.2 K the heat capacity of Ag(s) is 0.0145 J K^{-1} mol^{-1}. Assuming that the Debye law applies, determine $S_m(4.2\,K) - S_m(0)$ for silver.

Given

At **T = 4.2 K**, the molar heat capacity of silver is

$$C_m = 0.0145 \text{ J K}^{-1} \text{ mol}^{-1}$$

Debye law applies.

Debye law at low temperature

For solids at very low temperatures:

$$C_m = aT^3$$

So first find constant **a**:

$$a = \frac{C_m}{T^3} = \frac{0.0145}{(4.2)^3}$$

$$(4.2)^3 = 74.088$$

$$a = \frac{0.0145}{74.088} = 1.96 \times 10^{-4} \text{ J K}^{-4} \text{ mol}^{-1}$$

Entropy change

$$S_m(T) - S_m(0) = \int_0^T \frac{C_m}{T}\, dT$$

Substitute $C_m = aT^3$:

$$S_m(T) - S_m(0) = \int_0^T aT^2\, dT$$

$$= \frac{aT^3}{3}$$

Put values (T = 4.2 K)

$$S_m(4.2) - S_m(0) = \frac{1.96 \times 10^{-4} \times (4.2)^3}{3}$$

$$= \frac{1.96 \times 10^{-4} \times 74.088}{3}$$

$$= \frac{0.0145}{3}$$

$$= 4.83 \times 10^{-3} \text{ J K}^{-1} \text{ mol}^{-1}$$

Starting from the definitions of the thermodynamic potentials $H = U + pV$, $A = U - TS$, and $G = H - TS$, derive the remaining three Maxwell relations. Form the exact differentials of H, A, and G, substitute $dU = TdS - pdV$, and obtain the corresponding Maxwell relations. Hence write down the complete set of Maxwell relations.

1. Known starting point

The fundamental relation:

$$dU = T\,dS - p\,dV$$

with $U = U(S, V)$.

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2. Enthalpy $H = U + pV$

Differentiate:

$$dH = dU + p\,dV + V\,dp$$

Substitute dU:

$$dH = (TdS - pdV) + pdV + Vdp$$

$$dH = TdS + Vdp$$

So $H = H(S, p)$.
For $dH = MdS + Ndp$, with $M = T$ and $N = V$:

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

This is **Maxwell relation 2**.

3. Helmholtz free energy $A = U - TS$

Differentiate:

$$dA = dU - TdS - SdT$$

Substitute dU:

$$dA = (TdS - pdV) - TdS - SdT$$

$$dA = -SdT - pdV$$

So $A = A(T, V)$.
For $dA = MdT + NdV$, with $M = -S$ and $N = -p$:

$$\left(\frac{\partial(-S)}{\partial V}\right)_T = \left(\frac{\partial(-p)}{\partial T}\right)_V$$

or

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

4. Gibbs free energy $G = H - TS$

First, $H = U + pV$, but we already have $dH = TdS + Vdp$.
Differentiate G:

$$dG = dH - TdS - SdT$$

Substitute dH:

$$dG = (TdS + Vdp) - TdS -$$

$$dG = -SdT + Vdp$$

So $G = G(T, p)$.
For $dG = MdT + Ndp$, with $M = -S$ and $N = V$:

$$\left(\frac{\partial(-S)}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p$$

or

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

5. The first Maxwell relation

From $dU = TdS - pdV$, with $U = U(S, V)$:
$M = T$ and $N = -p$:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$